

*Are phonosyntactic rules necessary?**

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Selkirk (1986) proposes that rules of the phrasal phonology apply to a structure called P-structure that is derived from, but not isomorphic to, syntactic structure. While Selkirk claims that most rules of external sandhi fall into the category of P-structure rules, there are three rules that she suggests may still require direct reference to syntactic structure. These are Mende Consonant Mutation (Conteh *et al.* 1986), Kimatuumbi Vowel Shortening (Odden this volume), and 'perhaps even rules like English *wanna* contraction' (Selkirk 1986: 374). Selkirk says that these rules 'appear to be sensitive to a richer array of syntactic conditions than would be expressible via the translation of syntax into P-structure'.

In this paper we will re-examine the phonological status of two of these rules, Kimatuumbi Vowel Shortening and Mende Consonant Mutation. Kimatuumbi Vowel Shortening, discussed in §2, is shown to fall neatly into the category of P-structure rules as defined by Selkirk. Mende Consonant Mutation, discussed in §3, can also be treated as a P-structure rule but requires an elaboration of Selkirk's algorithm for constructing the phonological phrase.

1 Preliminaries

Selkirk (1986), following Chen (1985), sets forth a theory of prosodic domains. She claims that the algorithm building prosodic structure has access to well-defined levels of X' representations (X^0 , X^{head} and X^{max}). Each language specifies whether it uses the left or the right end of these categories in constructing prosodic domains. A prosodic domain is the span of material between these ends. The prosodic domain formed by reference to X^0 is the phonological word (ω), that formed by reference to X^{head} is the small phonological phrase (p), and that formed by reference to X^{max} is the phonological phrase (ϕ).

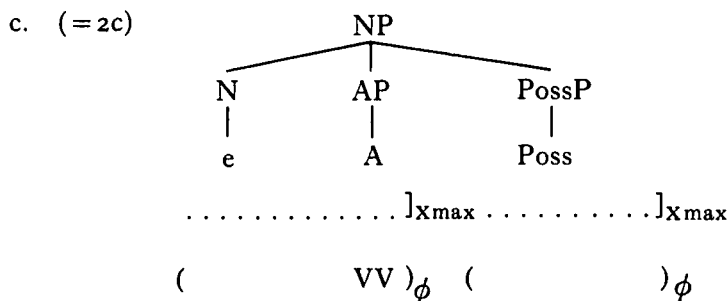
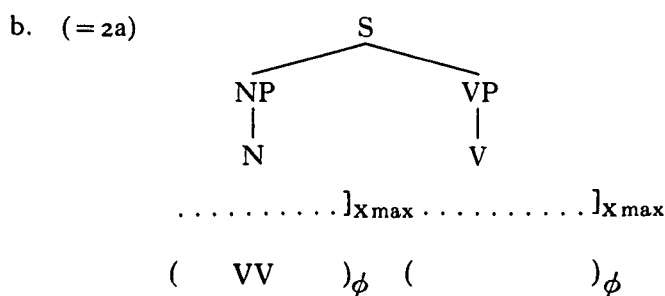
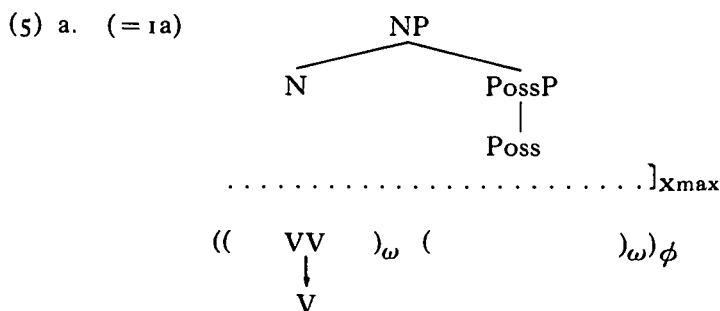
- (3) a. $[[\dots \underline{V} \dots]]_N [[\dots \underline{V} \dots]]_{\text{Poss}} [\quad]_{\text{NP}}]_{\text{PossP}}]_{\text{NP}}$
 kikolombe chake Mamboondo 'Mamboondo's cleaning
 shell'
 cf. kikoloombe 'cleaning shell'

- b. $[[\dots \underline{V} \dots]_N [[\dots \underline{V} \dots]_A []_A]_{AP}]_{NP}$
 mikate milaso milaaso 'long loaves'
 cf. mikaate 'loaf'
- c. $[[\dots \underline{V} \dots]_V [[\dots \underline{V} \dots]_N [Poss]_{PossP} []_{AP}]_{NP}]_{VP}$
 akakata kamba yaangu ngulu 'he cut my large rope'
 he-cut rope mine large
 cf. akakaata 'he cut'
 kaamba 'rope'

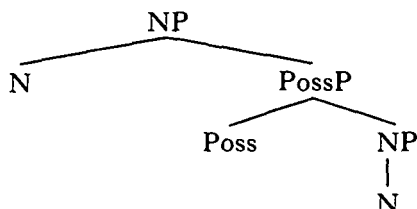
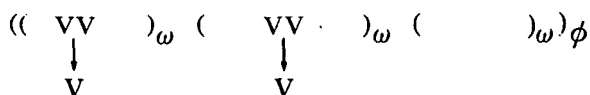
The rule of VS applies only to non-final words within a maximal projection, as can be seen from the structures in (1)–(3). In Selkirk's framework, this can be straightforwardly captured if the right edge of a maximal projection, $]_{X_{max}}$, defines the phonological phrase. The rule then applies between words within a phonological phrase, as formulated in (4):

$$(4) V: \rightarrow V / (\dots (\dots \text{---} \dots)_{\omega} (\dots)_{\omega} \dots)_{\phi}$$

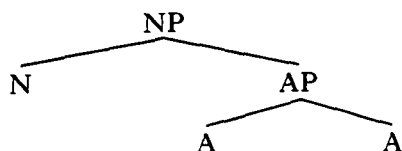
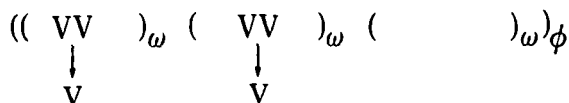
The application of VS is illustrated for sample forms in (5):



d. (= 3a)

.....]x_{max}

e. (= 3b)

.....]x_{max}

The rule applies within (5a) because the two words are in a single phonological phrase. VS fails to apply in (5b) and (5c) because the two words are not within the same phonological phrase. VS applies twice in (5d) and (5e) because there are three words within a phonological phrase. The rule can apply to the first two words.

A prosodic account of Kimatuumbi VS is straightforward. Given the success of this analysis, the other rules of Kimatuumbi with syntactic environments should be re-examined to see if they do in fact require the postulation of a phonosyntactic component of the grammar.

3 Mende

We have argued elsewhere (Conteh *et al.* 1986) that CONSONANT MUTATION (CM) in Mende must crucially refer to the traditional notion of c-command, where branching nodes determine the c-command domain. We argue here that if a single parameter, relevance of branchingness, is added to those proposed by Selkirk, Mende CM is not a rule of the phonosyntactic component but rather an external sandhi rule that applies within a phonological phrase.

The various environments in which mutation occurs are listed in (6). Underlining indicates that the consonant mutates:⁴

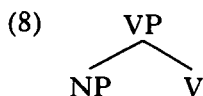
- (6) a. [NP V]_{VP} The initial consonant of the verb mutates
ì nyá wókòlóló ‘He imitated me’
he me imitate
cf. pókòlóló ‘imitate’
- b. [[NP N] Det]_{NP} The initial consonant of the head noun mutates
ndòpó ì hòwè í ‘the child’s knife’
child Det knife Det
cf. mbówé í ‘the knife’
- c. [N A]_{NP}⁵ The initial consonant of the adjective mutates
ngilà jèmbé í ná ‘that big dog’
dog big Det that
cf. sémbé ‘big’
- d. [NP P]_{PP} The initial consonant of the postposition mutates
ndèndé í bù ‘in the shade’
shade Det under
cf. mbù ‘under’
- e. [NP V]_S In intransitive sentences with a non-branching VP,
the initial consonant of the verb mutates
i. ndólàá wótèà ‘the baby turned’
baby turn
cf. pèté ‘turn’
ii. mú yèmbéíló ngúlí í hù ‘we swung on the tree’
we swung tree Det on
cf. fèmbé í ‘the swing’
- f. [P [N]_{NP}]_{PP} In prepositional phrases, the initial consonant of a
non-branching object noun phrase mutates
mèhê mé [à lòkó]_{NP} ‘eat with fingers’
food eat with hand
cf. tòkó ‘hand, forearm’
- g. [N N]_N⁶ In compounds, the initial consonant of the second
noun mutates
fèfè lëndèí ‘sailboat’
wind boat
cf. ndèndèí ‘boat’

CM fails to apply in the following environments:

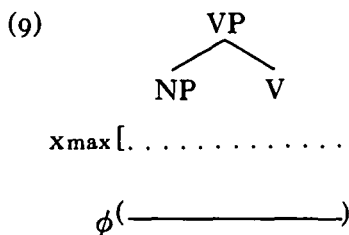
- (7) a. [e V]_{VP} If the object of a transitive verb is phonologically null, mutation fails to apply
 ndòpòi [e.c.] fèmbéngà 'The child swung it'
 child it swung
 cf. ndòpòi mbòmèi yèmbéngà 'The child swung the
 child hammock swung hammock'

- b. $[e P]_{PP}$ If the object of a postposition is phonologically null mutation fails to apply
 [e.c.] *mbù* 'under it'
 cf. *ndèndèi b̀ù* 'under the boat'
 boat under
- c. $[t V]_{VP}$ If the object of a transitive verb is extracted by a movement rule, mutation fails to apply to the verb
gbémiá ndòpòì t kpàndiá 'What has the child heated?'
 what child-Det heat
ngúlèí miá ndòpòì t kpàndiá 'It's the oil that the child heated'
 cf. *ndòpòì ngúlèí gbàndiá* 'The child heated the oil'
- d. $[NP [V PP]_{VP}]_S$ In intransitive sentences where the VP branches, the initial consonant of the verb fails to mutate
tí kàkpàngà ngì má 'they surrounded him'
 they surround him
 cf. **tí gàkpàngà ngì má*
- e. $[P [N Det]_{NP}]_{PP}$ In prepositional phrases where the object of the preposition branches, the initial consonant of the noun fails to mutate
hélé a ngúlí í 'hang from the tree'
 hang from tree Det

Much of Mende CM can be accounted for if the phonological phrase in Mende is defined by the left edge of a maximal projection, $x_{max}[$, and mutation applies between words within the phonological phrase. With this definition, the application of mutation in (6a, b, c, d, f) and its failure to apply in (7a, b, c, d, e) are accounted for. For instance, (6a) has the structure in (8):



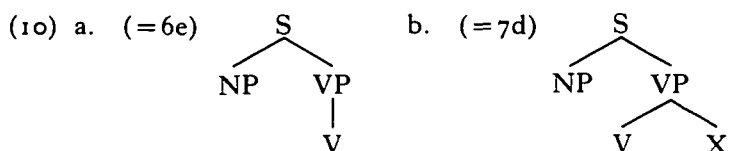
The phonological phrase formed by $x_{max}[$ is defined as in (9):



CM applies to the initial consonant of the verb, since it is preceded by another word within the phonological phrase. The other structures in (6)

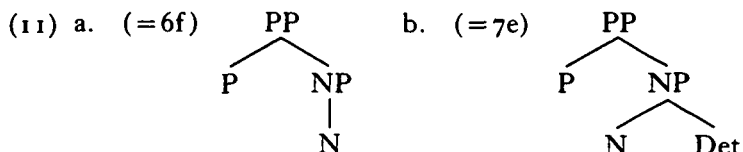
where CM applies are similar to (8). CM fails to apply in the examples in (7a, b, c) because of the presence of the empty category *e* or *t*. As discussed in note 4, an empty category does not trigger CM.

The intransitive verbs in (6e) and (7d) and the prepositional phrases in (6f) and (7e) require further comment. As shown in Conteh *et al.* (1986), in cases such as (6e), where the verb undergoes CM, the verb phrase does not branch. The structure of (6e) is given in (10a). In intransitive structures such as (7d), on the other hand, the verb phrase contains postverbal material that is subcategorised for by the verb. Thus, in this case, the verb phrase branches and CM between the subject and the verb fails to apply. The structure (7d) is shown in (10b):



where X is adverbial, postpositional or adjunct material in VP

The sentences with prepositional phrases are parallel to those with intransitive verbs. Structures for (6f) and (7e) are given in (11a) and (11b) respectively:



When the object of the preposition does not branch, as in (11a), CM applies; when it does branch, as in (11b), CM fails to apply.

Conteh *et al.* (1986) propose the following environment for CM:

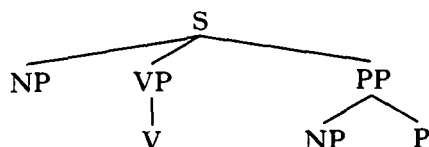
- (12) The initial consonant of Y mutates in [X][Y], where Y c-commands X.⁷

This rule correctly predicts that CM applies in the structures in (10a) and (11a) but not in those in (10b) and (11b). In (10a), the verb c-commands the subject NP, while in (10b) it does not. In (11a), the noun c-commands the postposition while in (11b) it does not. The phonological phrase structure derived using $x_{\max}$ alone assigns identical phrasing to the two structures.

The Mende data thus seem at first not to be amenable to analysis within the prosodic component of Selkirk's model. However, Mende can be accommodated prosodically if Selkirk's statement of the algorithm defining phonological phrase structure is slightly modified.

Selkirk proposes three well-defined domains of phonological structure. These are delimited by an edge of X^0 , X^{head} and X^{max} . We propose that Selkirk's 'maximal phonological phrase' may be defined either by X^{max} , as she proposes, or by a branching X^{max} ($X^{\text{max-b}}$). Thus in addition to the parameter specifying left or right edge, we propose a parameter of branchingness for the determination of the maximal phonological phrase. Mende chooses $X^{\text{max-b}}$ for this parameter, and specifies the left edge as the delimiter. The formation of the phonological phrase in Mende is illustrated in the examples in (13):

(13) a.⁸

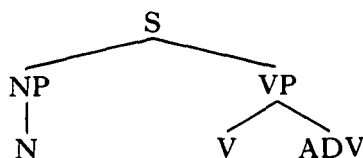


$x_{\text{max-b}}[\dots \dots \dots x_{\text{max-b}}[\dots \dots \dots$

$\phi(\text{—————}) \quad \phi(\text{—————})$

Mutation applies to V

b.



$x_{\text{max-b}}[\dots \dots \dots x_{\text{max-b}}[\dots \dots \dots$

$\phi(\text{—————}) \quad \phi(\text{—————})$

Mutation does not apply to V

Because the VP does not branch in (13a), it does not form its own phonological phrase. In (13b), where the VP branches, it is a phonological phrase on its own.

4 Conclusion

We have proposed to amend Selkirk's model of prosodic phonology by adding the possible relevance of branchingness at the level of the phonological phrase.⁹ Branchingness is by no means unknown in phonology. It is referred to at the prosodic levels of syllable and foot. While the branching parameter for defining the phonological phrase proposed in this paper

refers to the syntactic structure, phonological phrase construction is part of the phonological component of the grammar.

If the addition of reference to branchingness makes it possible to treat prosodically those processes which have been identified as phonosyntactic in nature, then this parameter is a small price to pay for the elimination of an entire class of rules.

NOTES

- * The authors are listed in alphabetical order; no priority of authorship is intended. We would like to thank Ellen Kaisse and Peter Avery for helpful comments.
- [1] The Kimatuumbi orthography is taken from Odden. For Mende, we use the standard orthography.
- [2] Examples (2c) and (2d) are discussed in detail by Odden. Because of data such as these, he concludes that only the head of a category undergoes shortening. In the analysis to be presented here, these data present no special problems.
- [3] Odden argues persuasively that the intensifiers are daughters of S rather than of VP.
- [4] The phonological mutations found in Consonant Mutation are examined in Cowper & Rice (1986). Specifically, we claim that CM involves a spreading of the features of a final vowel in the first word on to the initial consonant of the second word. CM cannot apply when a word is on an edge because there is no preceding vowel. It also cannot apply when a word follows an empty category (*e* or *t*) because these have no vowel from which features can spread.
- [5] We assume that adjectives are not maximal projections in Mende since they cannot be modified, they cannot take complements, and they cannot appear with degree specifiers. Such a position has been argued for for Ewe (Clements 1978; adopted by Kaisse 1985 and Selkirk 1986), for Xiamen (Chen 1985; Selkirk 1986), and for prenominal adjectives in French (Kaisse 1985; Selkirk 1986).
- [6] Notice that in (6g) CM applies between words within a word. Since we will claim that CM applies between words within a phonological phrase, this example is problematic. CM might also be a word-level rule, with 'word' replacing 'phonological phrase' within its structural description.
- [7] The definition of c-command relevant here is the traditional one involving branching nodes, rather than domain c-command, which refers to maximal projections. Domain c-command, as used by Kaisse (1985), will correctly block CM in (10b) and (11b), but will also wrongly block the rule in (10a) and (11a).
- [8] In (13a), a branching constituent is found following VP, and two phonological phrases are present. Our hypothesis predicts that if a non-branching constituent follows VP it cannot stand as a phonological phrase on its own and would be part of the same phonological phrase as the verb. All non-branching postverbal material invariably begins with a mutated consonant, and is thus consistent with our hypothesis.
- [9] This raises the empirical question of whether branching may be relevant at other levels of phrase phonology.

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